

Basic Principles of Medicine 1

Module: Foundations to

Medicine Lecture 19

Lecture Title: Use of Genomics in Diagnosis

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Pre-Lecture - Required Reading and Objectives

1. You can find the following pages and chapters on your Sakia site under:

- Book Access

FTM Genomics in Diagnosis Readings from Korf textbook which includes:

- Methods 1.1 – Isolation of DNA – p4
- Methods 2.2 – Polymerase chain reaction – p35
- Methods 4.1 – DNA Sequencing – p68
- Methods 4.3 – Gene Chips – p83
- Methods 4.4 – Next Generation Sequencing p84
- Hot Topic 4.1 – Genome Sequencing p 85 (note that this also defines the exome)
- Methods 6.1 – Chromosomal Analysis – p105
- Methods 6.3 – FISH – p108
- Methods 6.4 – Genomic Microarrays – P109
- Clinical Snapshot 6.3 – 22q11.2 Deletion Syndrome – p120
- Hot Topic 6.1 – Pathology Associated with CNV – p122
- Clinical Snapshot 4.1 Huntington Disease – pg74

2. DLA 17 now one recording and one set of notes

SOM.MK.III.BPM1.1.FTM.3.GNET.0069	Explain how triplet repeat expansion disorders are diagnosed
SOM.MK.III.BPM1.1.FTM.3.GNET.0070	Discuss why each triplet repeat expansion disorder needs its own unique test for diagnosis
SOM.MK.III.BPM1.1.FTM.3.GNET.0071	Explain how a G-band karyotype is prepared
SOM.MK.III.BPM1.1.FTM.3.GNET.0072	Describe FISH and how it can be used in diagnostic applications
SOM.MK.III.BPM1.1.FTM.3.GNET.0073	Compare and contrast : Metaphase FISH, Interphase FISH, and Chromosome painting (SKY FISH, or spectral karyotype)
SOM.MK.III.BPM1.1.FTM.3.GNET.0074	Describe the strategy behind the microarray: CGH Microarray, SNP Microarray, cDNA Microarray

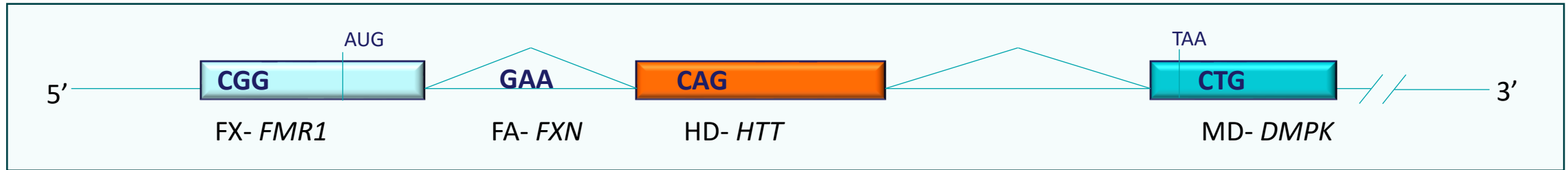
Objectives for Lecture & DLA: Genomics in diagnosis

SOM.MKIII.BPM1.1.FTM.3.GNET.0069	Explain how triplet repeat expansion disorders are diagnosed
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SOM.MKIII.BPM1.1.FTM.3.GNET.0075	Analyze genomic approaches that are used in the diagnosis of human disorders
SOM.MKIII.BPM1.1.FTM.3.GNET.0076	Analyze how PCR can be used to obtain a diagnosis of a genetic disorder
SOM.MKIII.BPM1.1.FTM.3.GNET.0077	Discuss how next generation sequencing is being used in genomic applications
SOM.MKIII.BPM1.1.FTM.3.GNET.0078	Describe the difference between whole genome sequencing (WGS), whole exome sequencing (WES), gene panels, and single targeted gene sequencing
SOM.MKIII.BPM1.1.FTM.3.GNET.0079	Describe the different types of mutations that may be observed in the human genome and explain how they are categorized.
SOM.MKIII.BPM1.1.FTM.3.GNET.0080	Consider the relative pros and cons of the genomic and genetic techniques described in this section, and their limits of resolution

Understanding the principles and applications of genetic testing techniques is essential for medical professionals. It enables accurate interpretation of test results and informed decision-making of patient care

Detection of Triplet Repeat Expansion Disorders

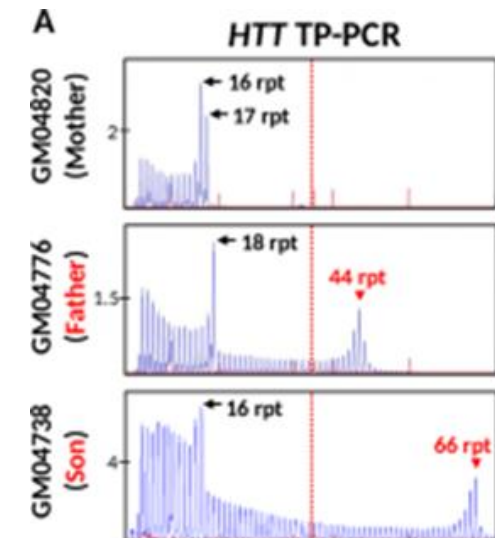
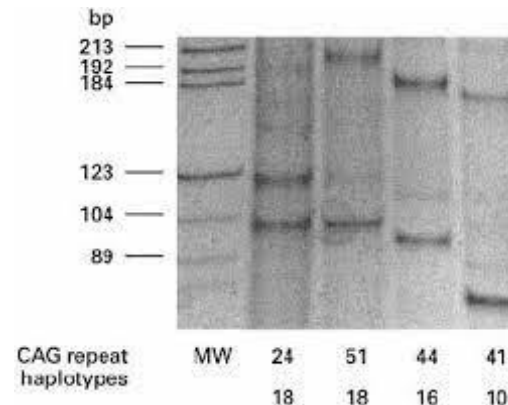
- Huntington disease (HD)- **autosomal dominant**
- Myotonic dystrophy type 1 (MD1)-**autosomal dominant**
- Fragile X syndrome (FX)- **X linked**
- Friedrich ataxia (FA)- **autosomal recessive**



- Only detected by PCR amplification of the area and sizing:

1. Electrophoresis after PCR
2. Column chromatography after PCR

Each disorder must have its own unique primers specific for gene and region of mutation



Other uses of PCR in Diagnosis

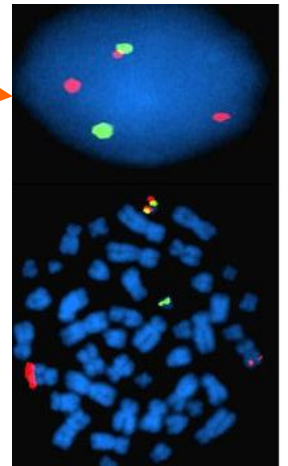
- Forensic analysis
- Viral, bacterial and fungal infections
- If targeted to a specific region, DNA can be amplified for sequencing, RFLP analysis, forensic analysis, prenatal diagnosis

Direct Methods 1: Hypothesis driven

- Make a probe to hybridize your target
 - Southern blot, Southern-RFLP- ssDNA probe 300-1000 bp
 - ASO blot- ssDNA probe 15-30 bp
 - PCR-RFLP, PCR sizing- ssDNA probes are a pair of primers 15-30 bp
 - FISH- ssDNA probes 10,000 to 20,000 bp
 - Northern- ssRNA or ssDNA about 100 bp
 - Western- Antibody
 - Sanger sequencing- ssDNA primer 15-30 bp

Interphase FISH

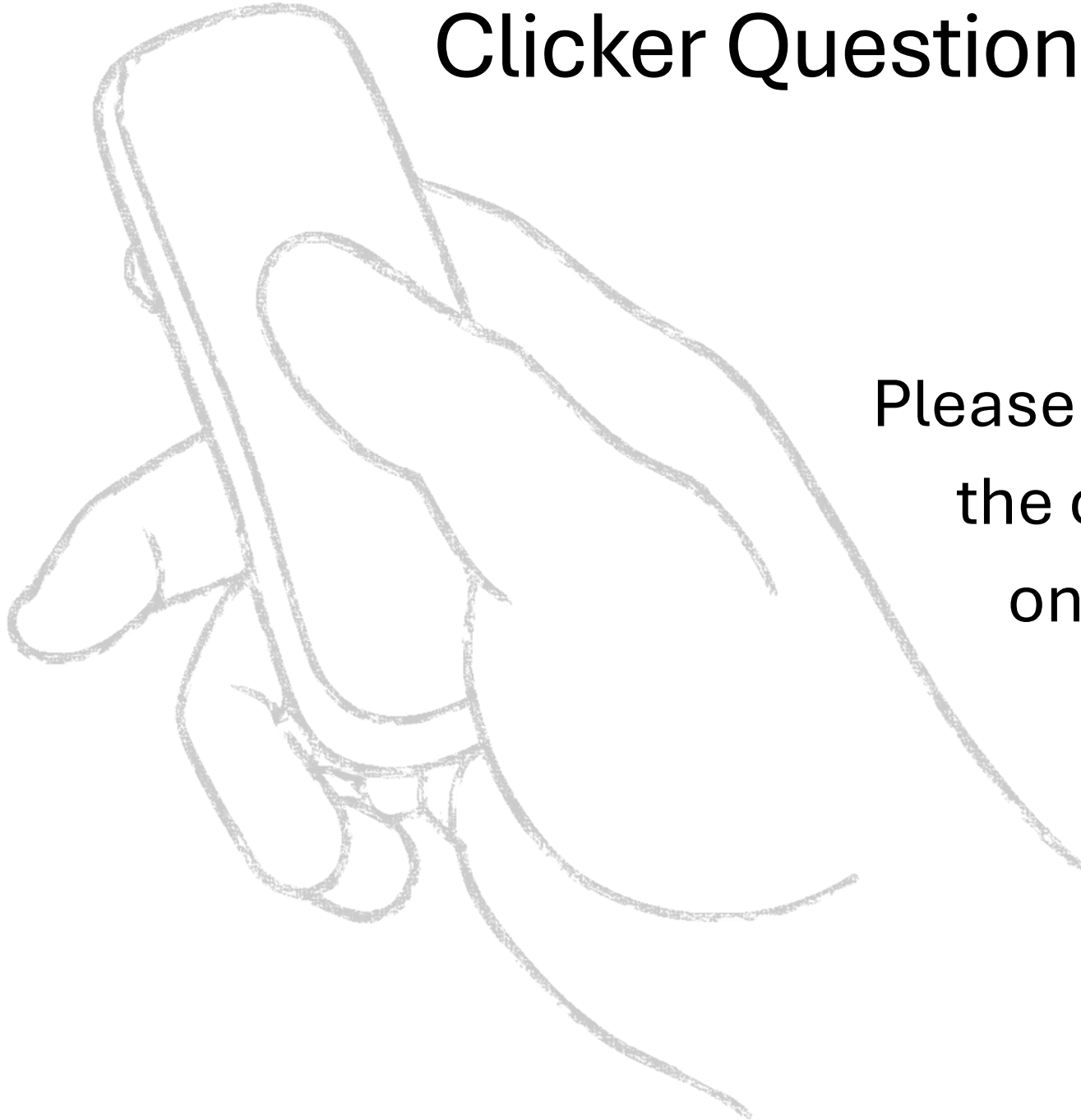
Metaphase FISH



Direct Methods 2: Query Entire Genome

- **Karyotype:** G-banding
- **Microarray:** Comparative Genome Hybridization (CGH), Single Nucleotide Polymorphism (SNP), Expression Array
- **Spectral Karyotype** (SKY, paint each chromosome a different color)
- **Next Generation Sequencing:** obtaining sequence in multiplex fashion
 - Whole Genome Sequencing (WGS)
 - Exomes of known genes (whole exome sequencing, WES)
 - A panel of related genes (e.g., all known genes involved in cardiomyopathy)
 - One gene with in-depth read (sequence one gene about 50 times)

Clicker Question Next!

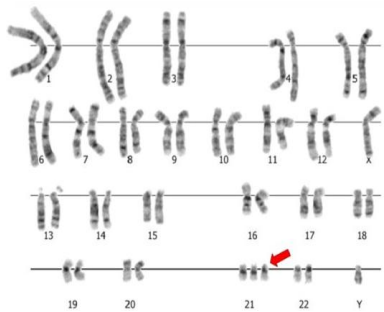


Please prepare to answer
the clicker question
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Link this concept: Allelic vs. Locus Heterogeneity

Allelic Heterogeneity		Locus Heterogeneity	
Same gene		Different genes	
Different mutations on same gene		Different genes have mutations leading to same syndrome	
Example: Cystic Fibrosis: AR inheritance		Example: Treacher Collins: AR or AD inheritance	
CFTR gene: Common mutations shown 		Mutations leading to non-functional proteins TCOF1 on chromosome 5 POLR1C on chromosome 6 POLR1D on chromosome 13	
CFTR Mutation F508 CFTR Mutation p.G970D CFTR Mutation c.3120 +1G>A CFTR Mutation p.R117Hmany others	Severe phenotype Moderate-severe phenotype Moderate-severe phenotype Mild phenotype	TCOF1 POLR1D POLR1C	Treacher Collins – AD Treacher Collins – AD Treacher Collins – AR

Genomic Tests- All about the resolution

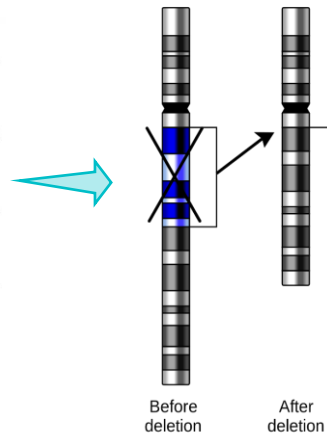


G-banding

Whole chromosome aneuploidy

Trisomy 21

Count: 47 chromosomes

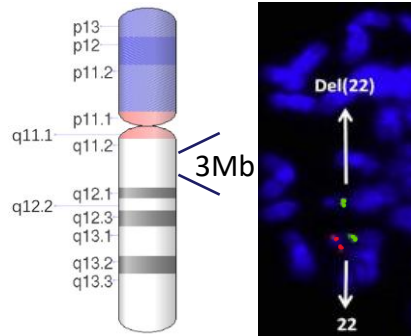


G-banding

deletions or duplications if banding pattern is affected

>5 million base pairs

(5Mb)



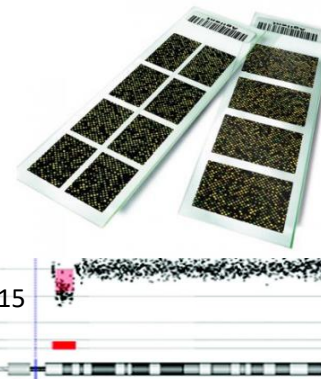
FISH

Deletions, limits depend on size of probe
10-1000 Kb to detect deletions
>0.5 million base pairs (0.5Mb)

Microdeletion disorders:

DiGeorge (137 kb)
Prader-Willi (125 kb)
Angelman (180 kb)
Williams (215 kb)
Wolf-Hirschhorn (98 kb)
Cri du chat (561 kb)

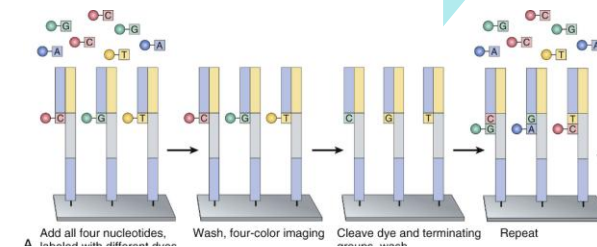
Some cancers



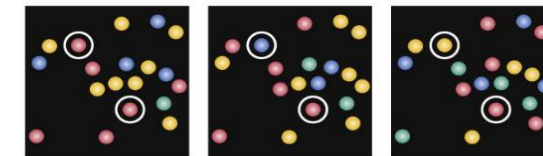
Array CGH

Euchromatic Genome
Resolution depends on how dense the chip is,
244K and 400K =
244,000 probes or
400,000 probes

Would give space between the probes as 8.9 kb and 5.3 kb respectively



C
A
T
G



Next Generation Sequencing

Euchromatic Genome
one base pair at a time

“Sequencing by synthesis”

Resolution depends on
What you are querying
by reference library

Whole Genome (WGS)
Whole Exome (WES)
Panel of candidate genes
One gene: in-depth read
(50-100x)

Karyotype Analysis

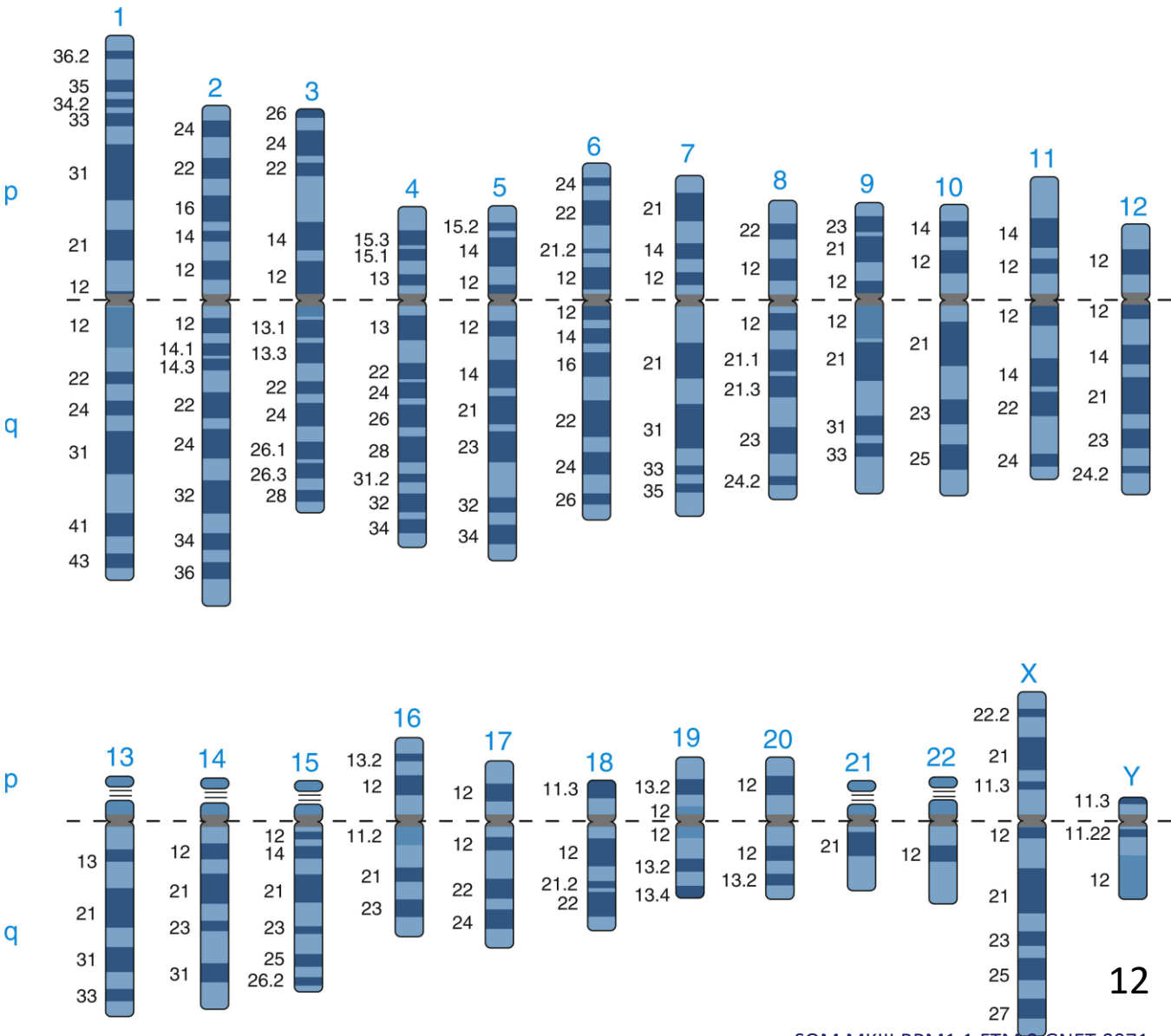
- Developed in 1950's
 - Prepared from lymphocyte cell culture arrested in metaphase, fixed to a slide and stained with Geimsa stain (AT rich regions)
 - Chromosomes from one cell called Chromosome or Metaphase Spread
- Can see chromosome number, large translocations, large insertions and deletions (gains or losses)
- Rule of thumb: deletions or duplications >5Mb are usually seen
- Deletions or duplications too small to be seen are <5Mb called “submicroscopic deletions” or “microdeletions”
- Relatively low resolution, changes must affect the banding pattern

Karyotype: Dr. Maj (SGU Grenada)

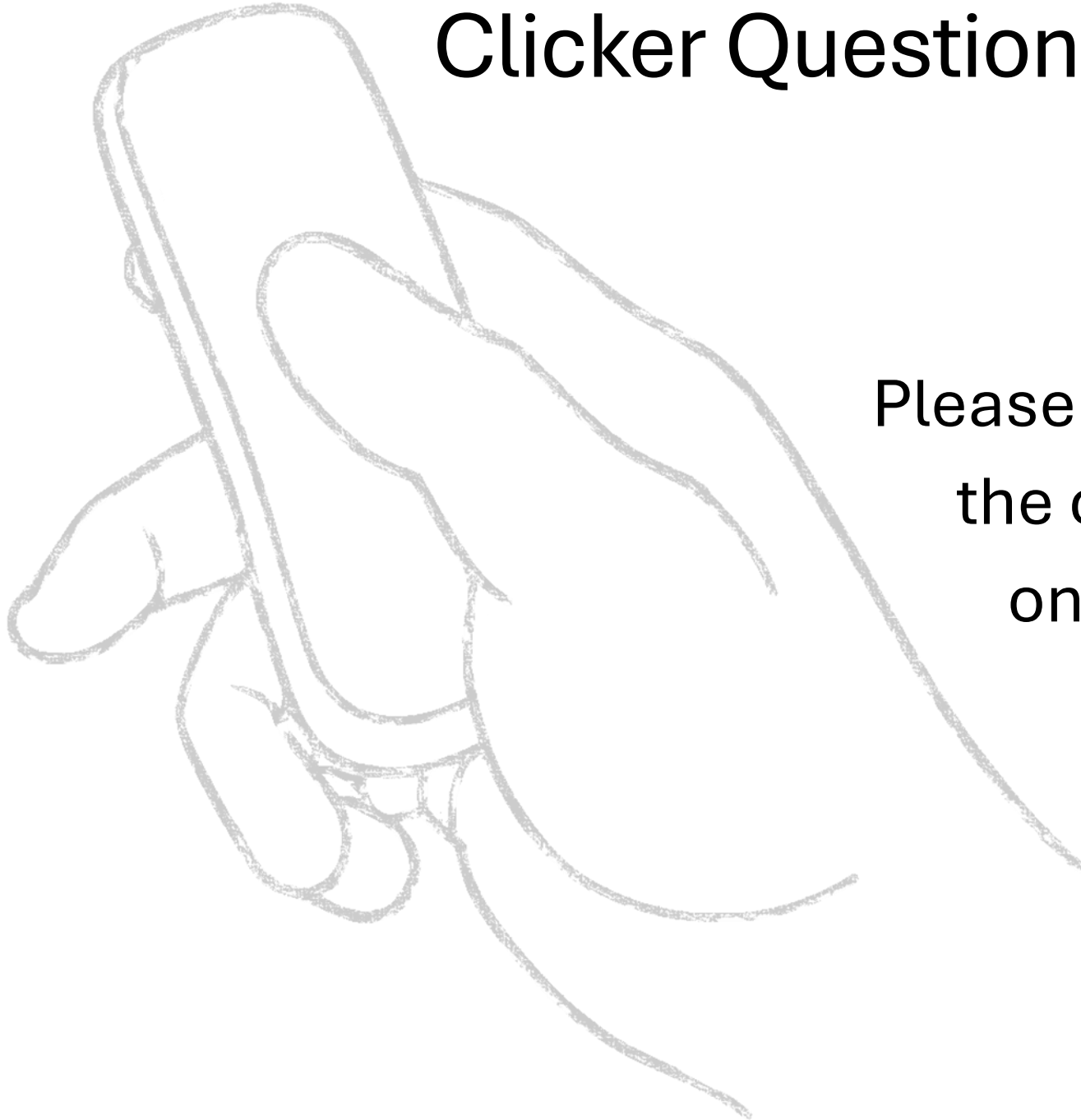


46, XX

Normal Karyotype, all
banding patterns seen



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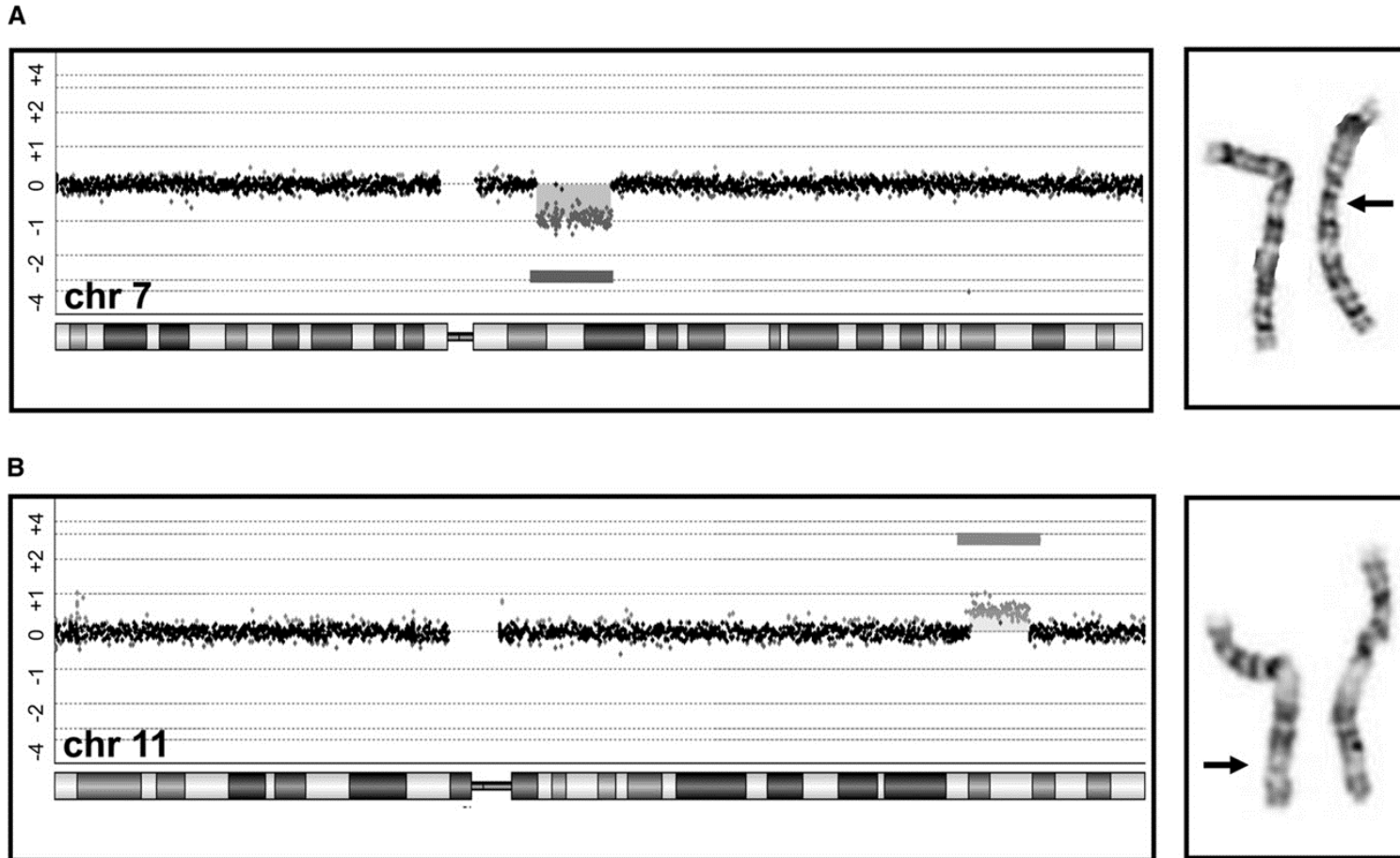


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Array Comparative Genome Hybridization (CGH)

- Looks for **copy number variants** (CNV) in patients compared to control
 - Gains or losses
- We have two copies of most of our genes, **humans are biallelic**
 - (exceptions *AMY1*, *CCL3L1C* but there are others)
- Replacing Karyotype as a first-tier test for patients with:
 - Developmental delay
 - Congenital anomalies
- Better resolution than G-banding
- Queries whole genome:
 - FISH looks for one microdeletion per test

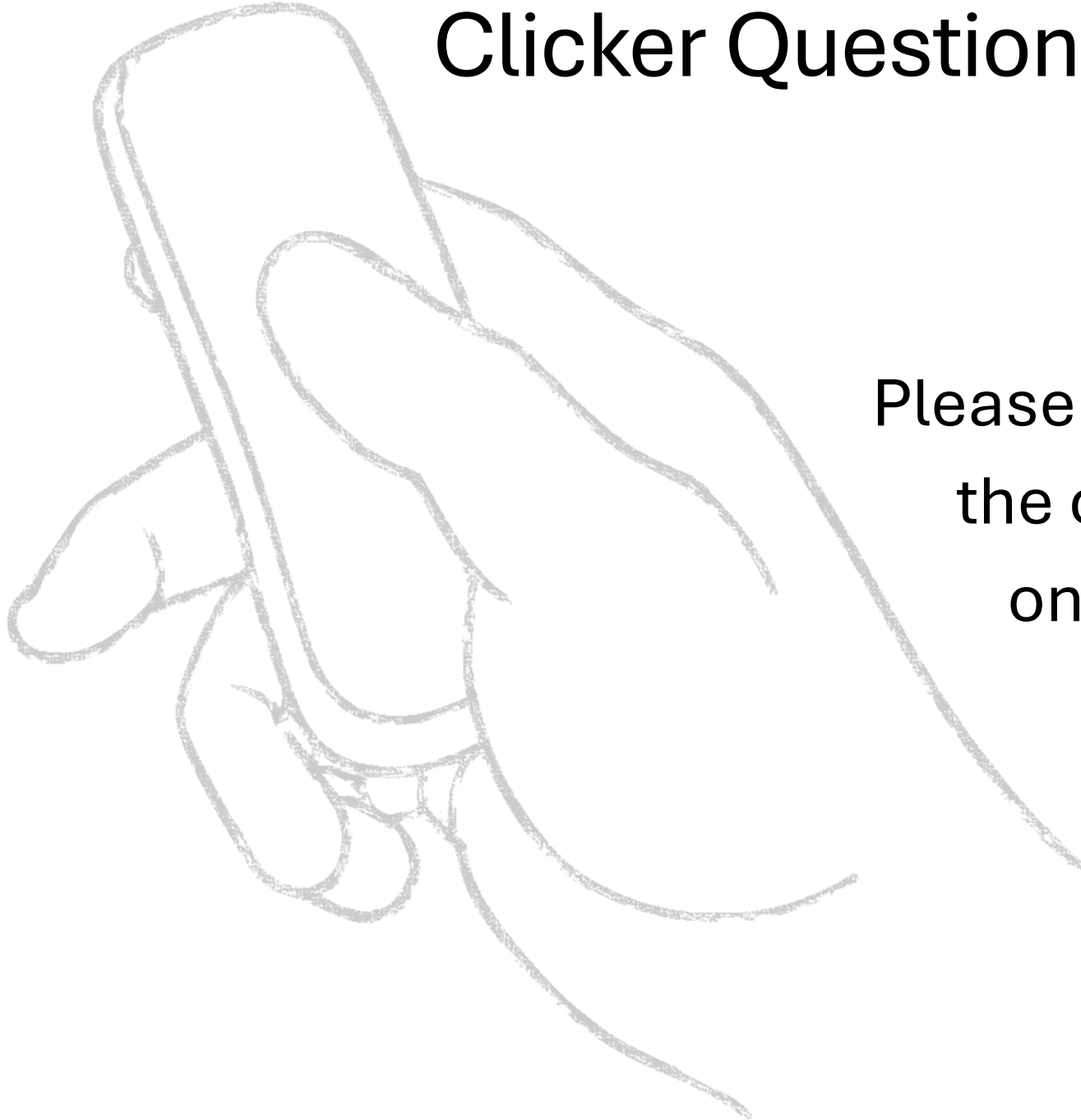
CNV that looks normal on Karyotype



- Chr. 7 >10 Mb deletion, not seen by karyotype

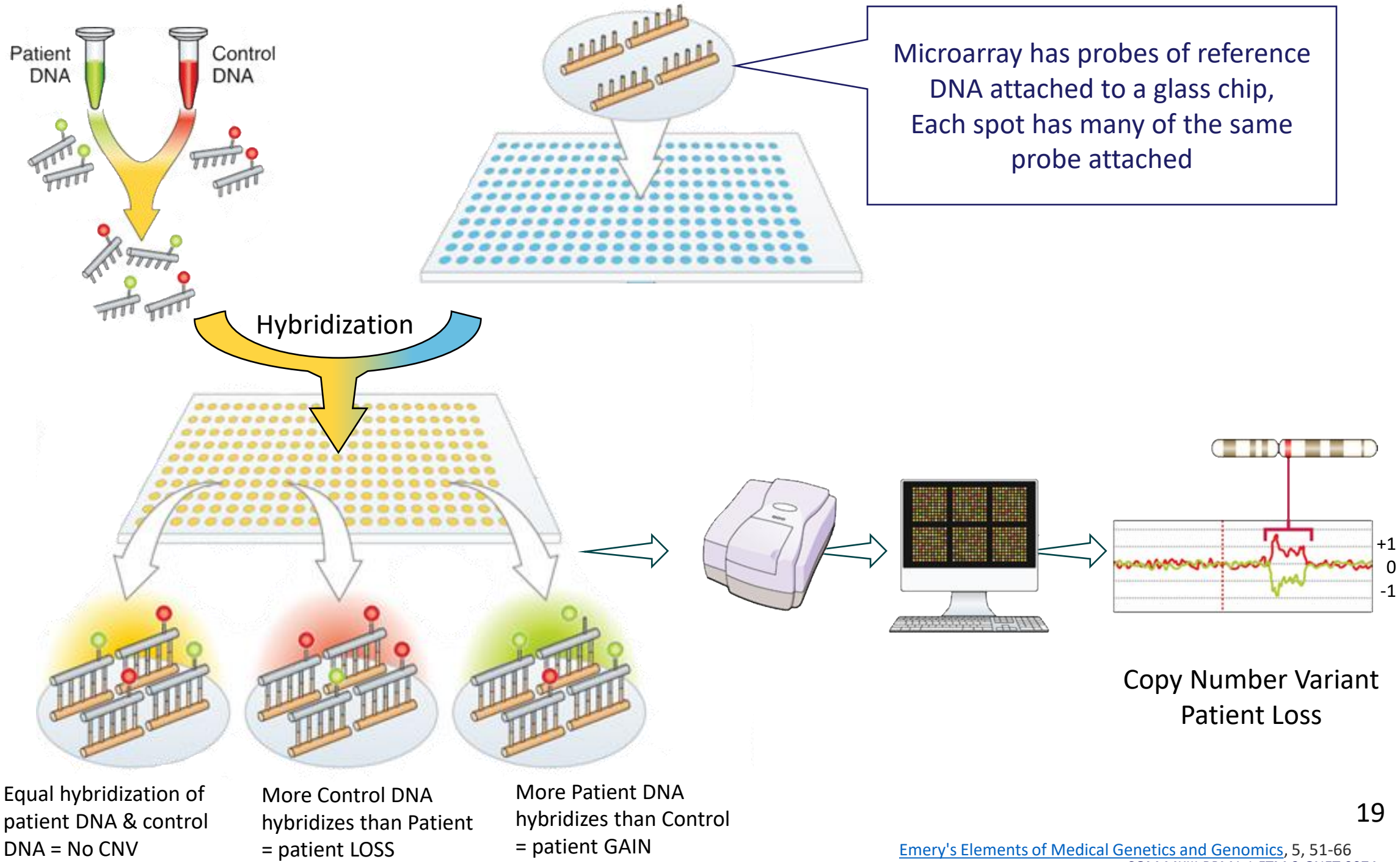
- Chr. 11 >7 Mb duplication, not seen by karyotype

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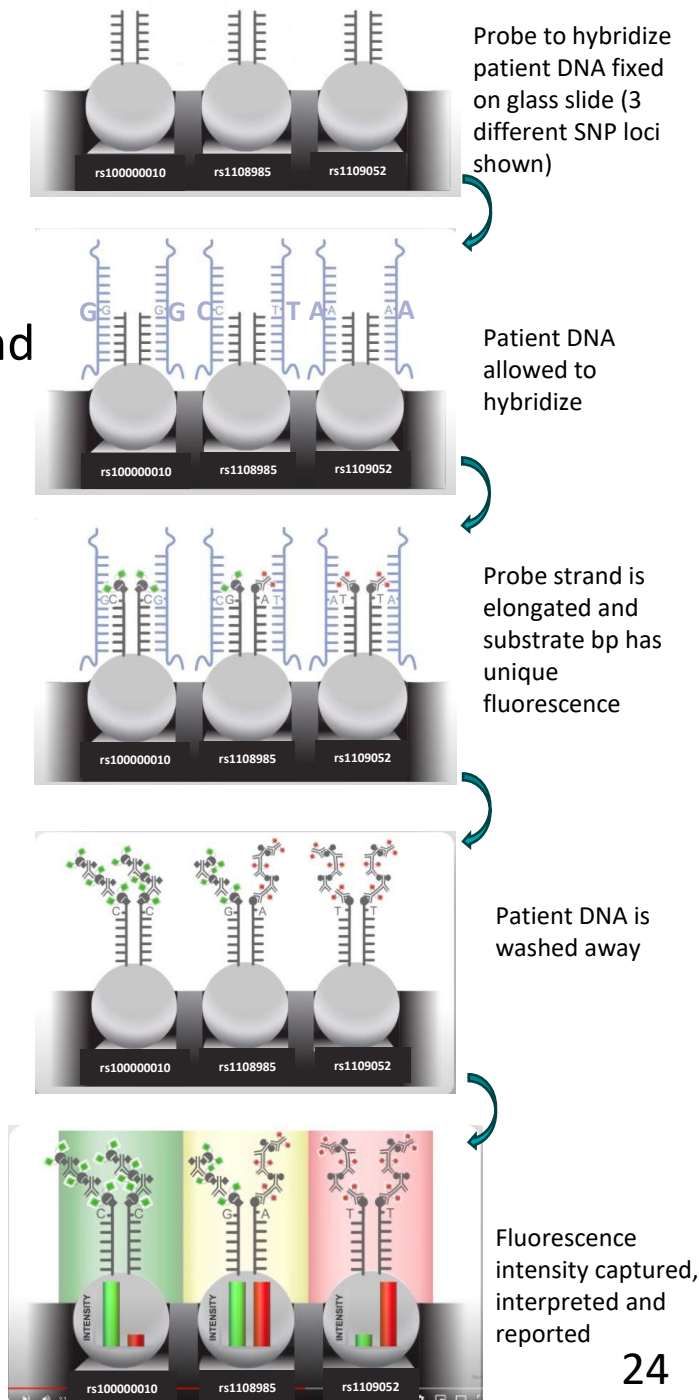
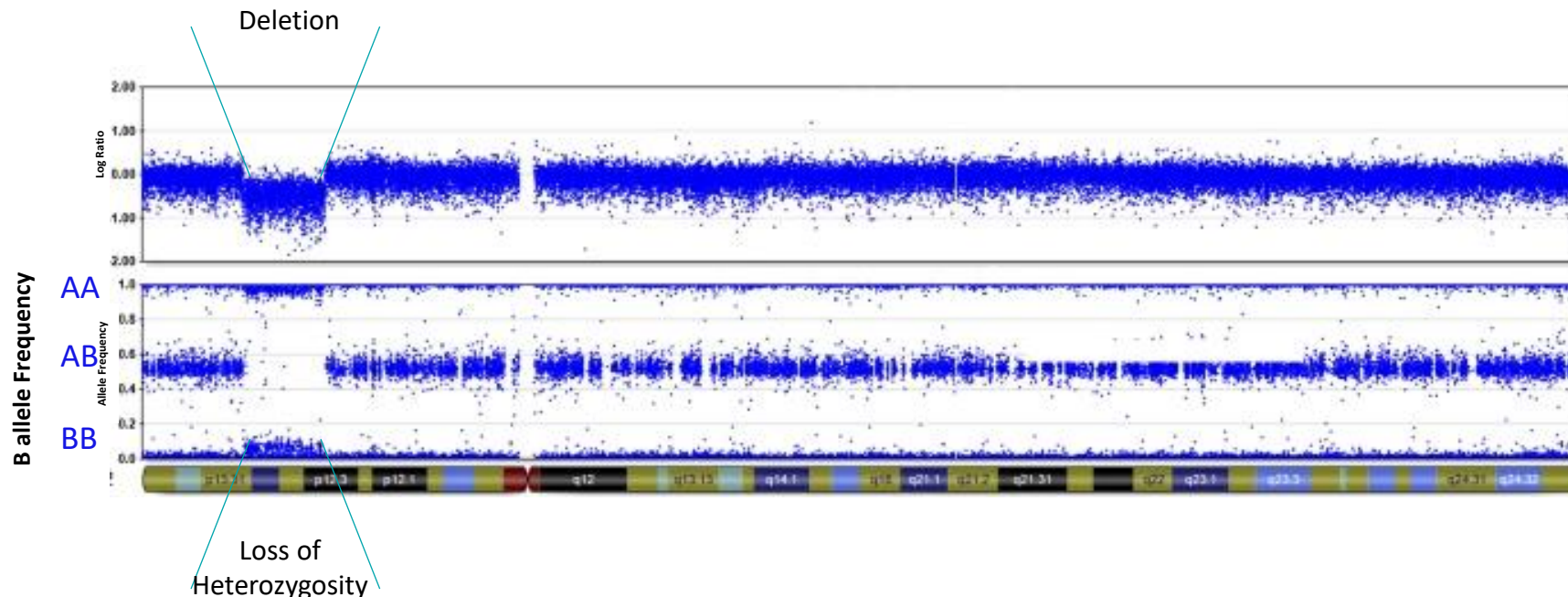
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DNA is sheared and a fluorescent probe attached



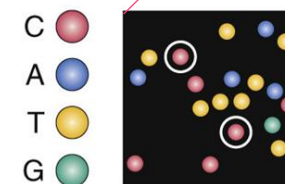
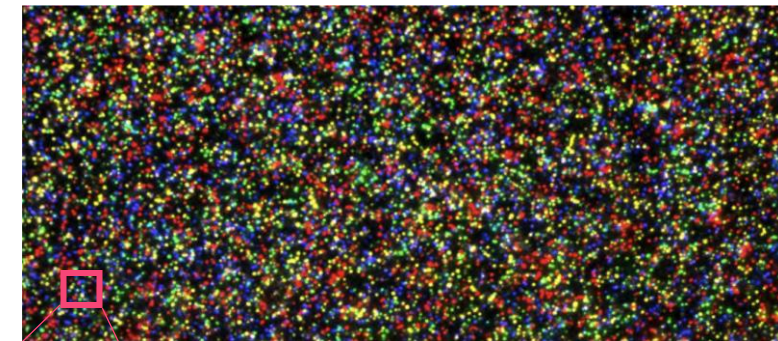
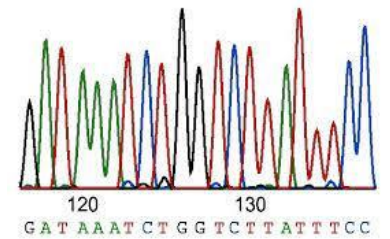
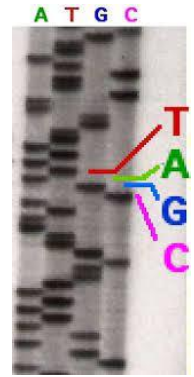
SNP Microarray Design: current tech

- Selected Polymorphisms by *array manufacturer*:
 - To have A allele or B allele
 - Will show 3 clusters for majority of people AA & AB & BB
- Probe=strands of DNA fixed on a slide (hundred thousand to millions of SNP's) and probe stops just below the target polymorphism
- Patient DNA is fragmented and hybridized to chip
- Probe DNA strand is elongated one base pair that has a fluorescent tag
- Fluorescence intensity is captured, and genotype assigned
- Loss of heterozygosity = deletion or uniparental disomy of a whole chromosome



DNA Sequencing

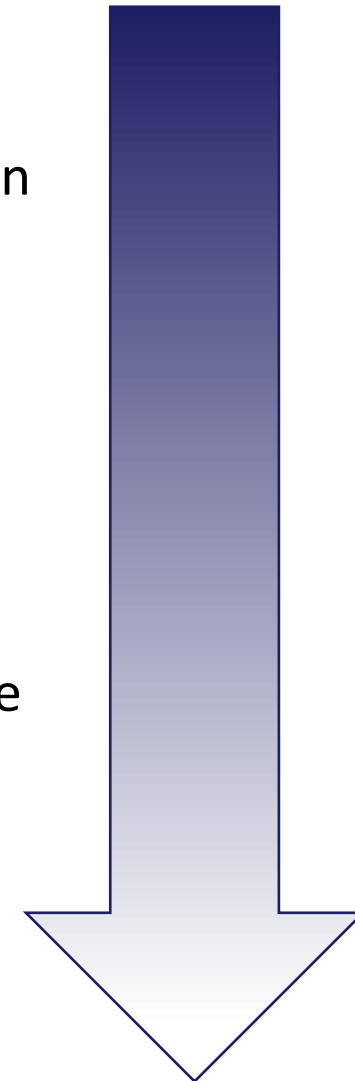
- Sanger sequencing using P-32 labeled nucleotides
 - Dideoxy chain termination, radioactive, manual, slow, ~1000 bp at a time
 - 4 tubes required to obtain one sequence information
- Automated sequencing using fluorescent dideoxy nucleotides
 - Each base a different color for automation, ~1500 bp at a time
 - Each sequence reaction is one tube for one sequencing reaction
- Next Generation Sequencing (NGS)
 - Fluorescence, lasers, computer analysis
 - Massively parallel obtainment of sequence data fragments
 - Need reference map, the human genome, to align sequence data fragments
- Next, next, next generation
 - New technologies continue to be discovered



Next Generation Sequencing (NGS)

- **Whole Genome Sequencing (WGS)**
 - All 6 billion base pairs (we are diploid)
 - Terabytes of information is processed, many variants of unknown significance found (both coding and non-coding genetic information)
- **Whole Exome Sequencing (WES)**
 - 1-2% of genome that encodes for proteins
 - Exons plus intron/exon boundary information (acceptor and donor splice site information)
- **Targeted Gene Panels**
 - A group of genes involved in a common pathway or disease state (pathology)
 - E.g. gene panel for cardiomyopathy
- **Single Gene Analysis**
 - In depth analysis of a single gene
 - Can identify heterozygosity

Relative cost and
effort to analyze
data

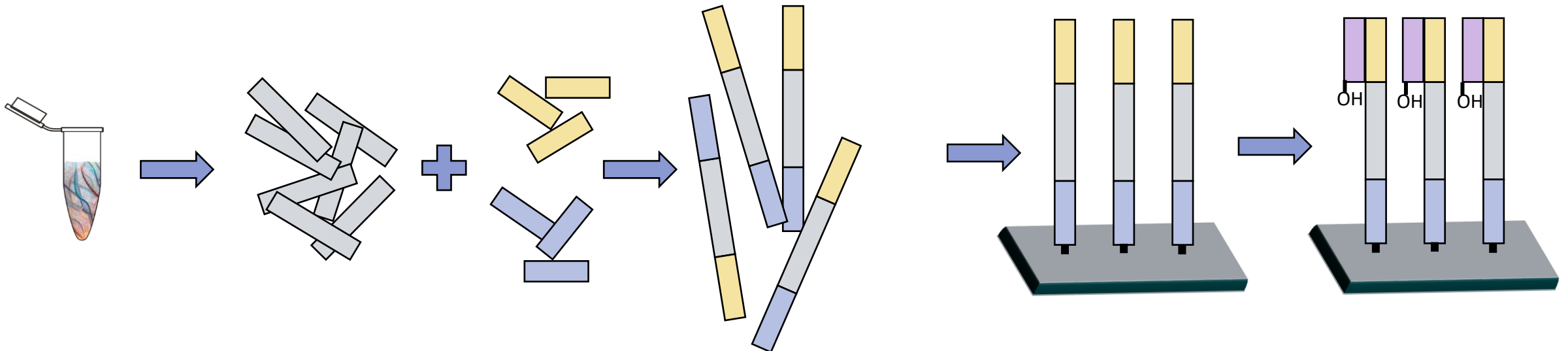


More data, more
analysis, more
variants identified
and more \$\$\$\$

Less data, less
analysis, less variants
found & less \$

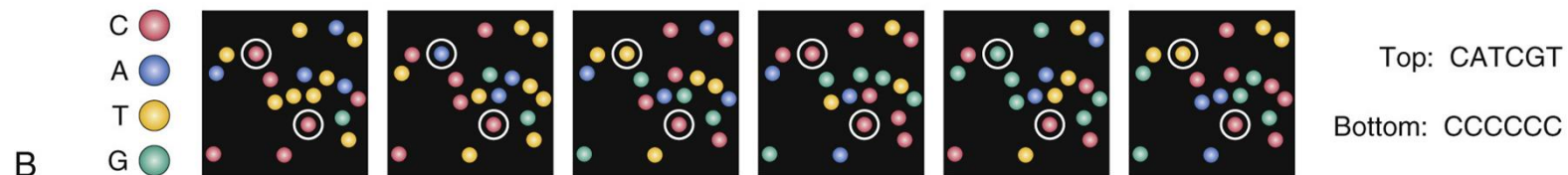
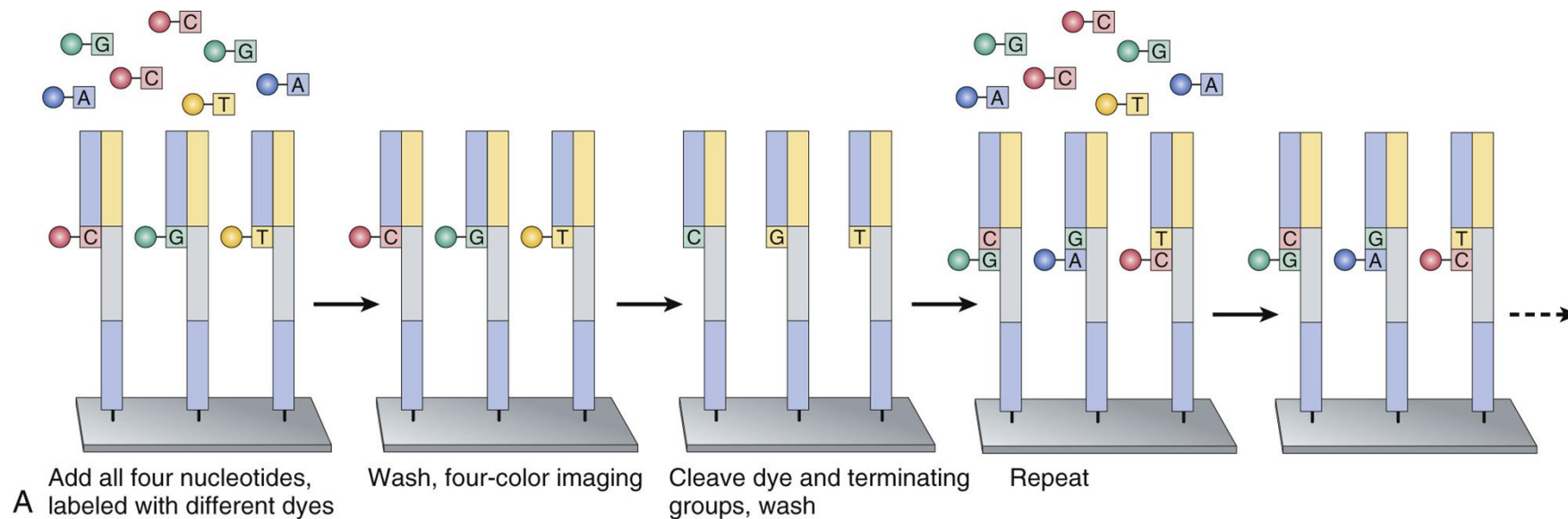
NGS: Massively parallel sequencing on a chip

1. DNA is isolated from patient cells
2. Break DNA into fragments of similar sizes
3. Add adapters to the DNA fragments
4. Attach DNA fragments to the surface of flow channels
5. Attach DNA primer with 3' OH group



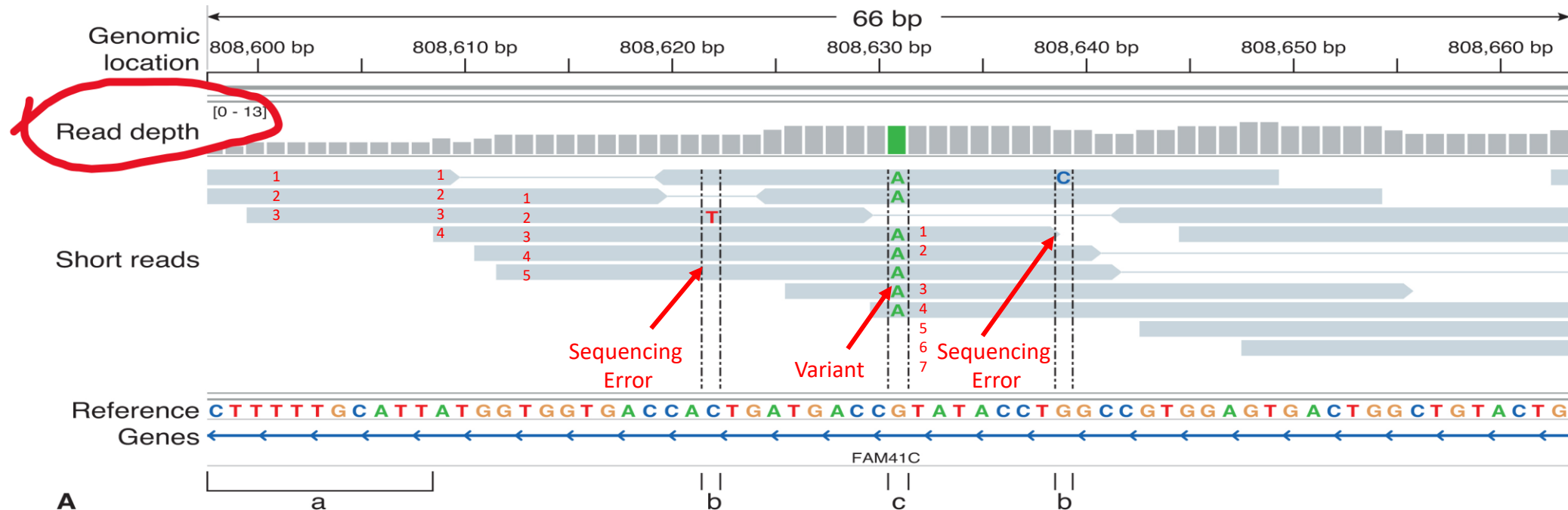
NGS: Fluorescent Bases Added One at a Time

1. Fluorescently labelled base pairs are substrate for synthesis reactions
 - One base pair is incorporated into each growing chain
2. Color imaging is taken & fluorescent tag is cleaved
3. Process is repeated & computer collects data for each fragment



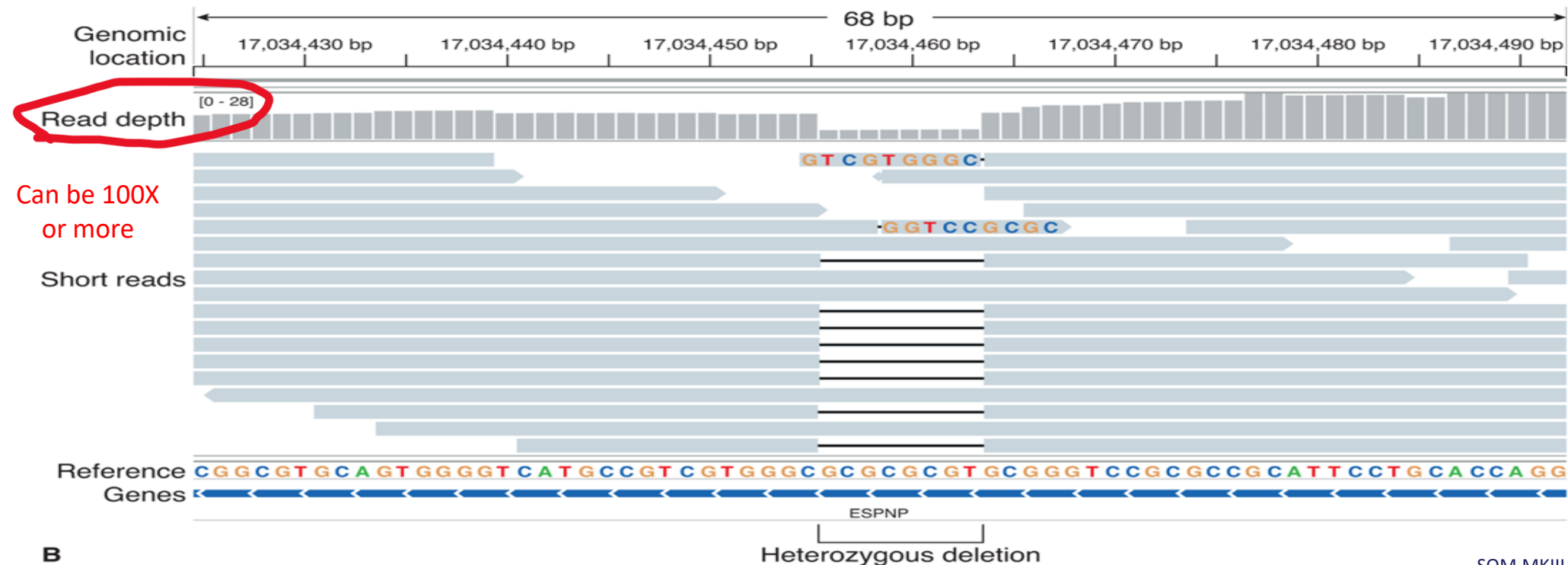
NGS: Sequence Information from Fragments Aligned to Reference Sequence

- Alignment of sequence fragments compared to reference genome
- Sequence is determined
- Read depth is a measure of how many sequences were compared to reference



NGS: In Depth Read

- Accuracy of sequencing is improved with more reads
- Allows for the determination of heterozygosity with better confidence
- Multiplex or massively parallel sequence acquisition: millions of sequences obtained simultaneously



NGS: Variants of Unknown Significance (VUS)

- May or may not be disease causing
 - Polymorphism if seen in people who do not have the disorder
- Sequencing may show previously unidentified variant in DNA (rare)
- A rare variant can be predicted to be disease causing
 - Predicted loss of function
 - Frame shift, non-sense, splice site, premature stop codon
- Variant may be in non-coding DNA region with unknown significance
- Sequencing whole genome will typically show many VUS's
- Sequencing whole exomes will show less VUS's
- Sequencing a panel of genes, even less VUS's

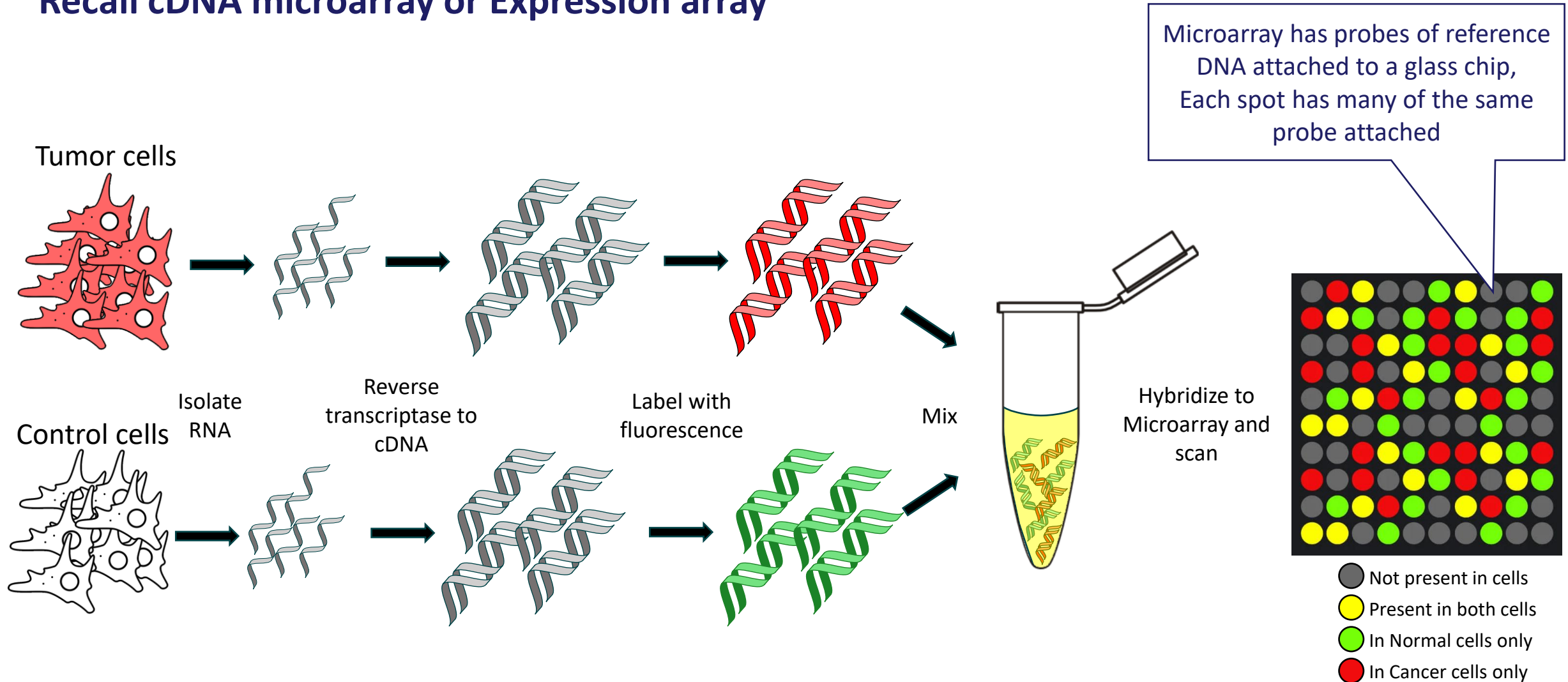
NGS: Gene Panels- Obtaining sequence data from targeted selection of genes

- **Targeted Panels:** During DNA preparation, DNA from the genes on panel are enriched through hybridization or PCR amplification
- **Virtual Panels:** Sequencing of genome performed, only relevant genes are analyzed
- **Types of Panels (either targeted or virtual)**
 - Cardiovascular health
 - Unexplained epilepsy
 - Endocrine disorders
 - Intellectual disability
 - Cancer panels
 - Fusion genes (~21 genes)
 - Insertions/deletions (~35 genes)
 - Gene amplification (~19 genes)
 - Myeloid panel

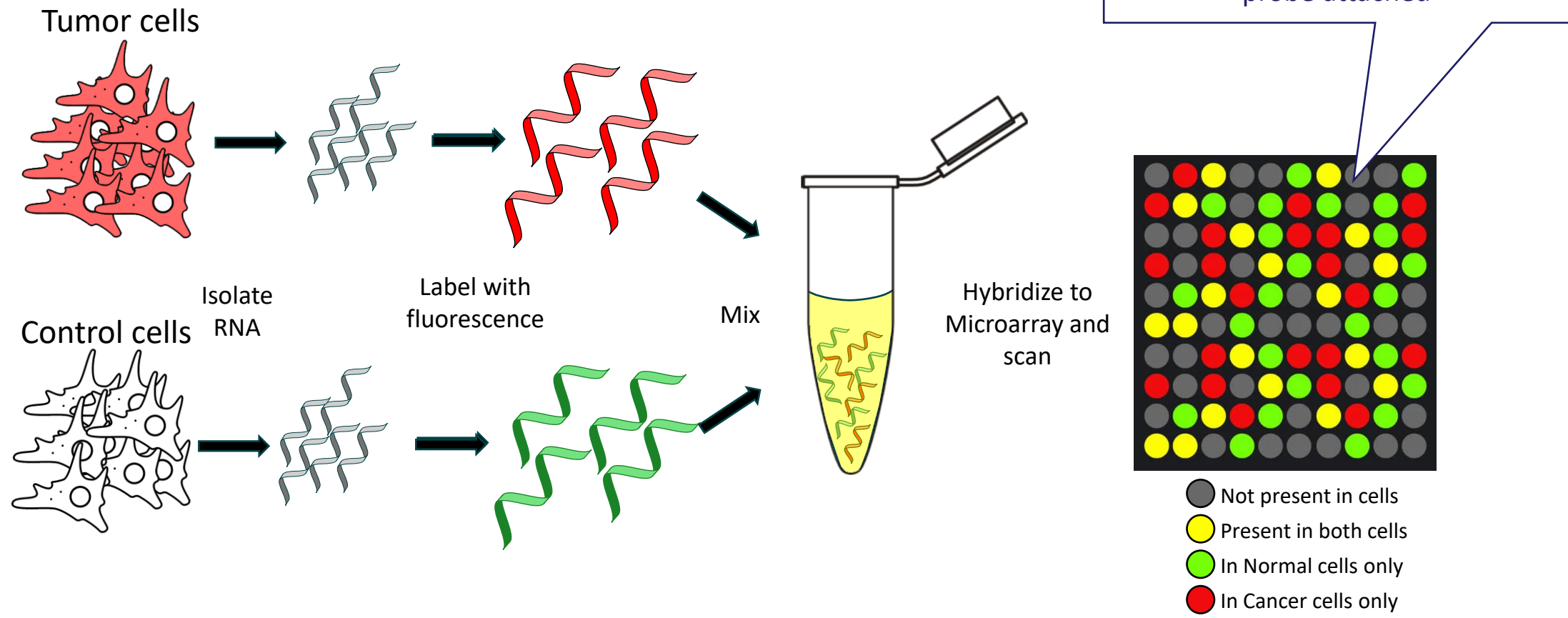
Cancer Gene Panels with cDNA or RNA

- Use of “transcriptomics”
- Analyze gene expression (mRNA) of a set panel of genes involved in cancer
- Can give numerical score that predicts prognosis or response to treatment etc.
- Relies on:
 - The cDNA microarray (expression array)
 - RNAseq (NGS for RNA)

Recall cDNA microarray or Expression array



Recall RNA microarray or Expression array



Array CGH is becoming a first-tier clinical diagnostic test for suspected developmental disabilities/congenital anomalies (CNV) vs. G-banding

Exome Sequencing is becoming a first-tier clinical diagnostic test for suspected neurodevelopmental disorders

Summary of Genomic Testing vs. Genetic Testing

Query entire genome:

- G-band
- SKY FISH
- Array CGH
- SNP Chip
- cDNA microarray
- NGS (WGS, WES)

Query a specific target

(you need to make an oligonucleotide/s specific as a probe or primers)

- ASO blot
- RFLP
- PCR
- Southern blot
- FISH interphase & metaphase (locus specific)
- Sanger sequencing (sequencing a single gene)

Query a specific target

(isolate data collection or enrich DNA sample)

- NGS (panel or in-depth read of one gene)

Summary of Genomic Tests

Test	Technique	Diagnosing genetic disorders
Next Generation sequencing	A high-throughput technique that sequences millions of DNA or cDNA fragments simultaneously. It allows for the detection of genetic variations compared to a reference genome. RNA is now being sequenced but with other type of technology which includes nanopores	Whole genome sequencing (WGS) determines complete DNA sequence for both coding and non-coding regions Whole exome sequencing (WES) focuses on sequencing protein coding regions cDNA sequencing cDNA is made from mRNA and gives information regarding expression
G-banding	Patient lymphocytes are cultured (grown in media) and arrested in metaphase by the addition of a spindle inhibitor like colchicine. Chromosomes are fixed to slide and dyed with Giemsa which produces characteristic banding patterns. (AT-rich regions stain deep purple)	Analysis of chromosomes to detect chromosomal abnormalities like Triploidy, large deletions, duplications, translocations and inversions. Resolution, must be larger than > 5 Mb, or must change banding pattern. Can be combined with FISH if a microdeletion is suspected
Microarray	A method that allows for the simultaneous analysis of thousands of DNA sequences. It involves hybridizing DNA samples to a solid support containing immobilized probes. Microarrays are used for gene expression profiling, genotyping, and detecting copy number variations.	CGH – copy number variants (duplications or deletions but never useful for translocations) SNP – genotyping/haplotyping, UPD (when comparing to both parents) Can give information on duplications and deletions when a loss of heterozygosity is identified Expression – compare gene expression between to populations of cells (different tissue, diseased vs. normal tissue in same organ) can be RNA or cDNA
Fluorescence In Situ Hybridization (FISH)	A technique that uses fluorescent probes to detect specific DNA sequences on chromosomes. FISH is commonly used to identify chromosomal abnormalities, such as gene rearrangements or aneuploidies.	Metaphase – performed on cells that have been cultured and arrested in metaphase, can be performed on same samples prepared for G-banding Interphase – performed on cells fixed to a slide with no need for culturing, sometimes referred to as Rapid FISH SKY FISH – thousands of probes designed to color each chromosome in a particular color, best to see cancer rearrangements, not informative for duplications or deletions, can be metaphase or interphase DNA
Polymerase chain reaction (not a genomic test but in this lecture)	A technique used to amplify a specific DNA sequence. It involves cycles of denaturation, annealing, and extension using DNA polymerase.	Triplet expansion repeats, VNTR or STR fingerprinting, deletions by PCR sizing, amplifying regions of DNA for further testing

Some specific points for students to use as study guide: Genetic & Genomic Tests

- Can you differentiate between indirect and direct methods of mutation detection?
- Can you outline the laboratory techniques, analyze, compare and contrast, and interpret results of some of the common tests that are used for molecular diagnosis of mutations
- Can you discuss how sizing of PCR products may be useful in genetic diagnosis
 - Use of PCR of regions that have variable amounts of repeat expansions (such as VNTR and STR – use in gene mapping, forensics, and paternity testing)
 - diagnosis of triplet repeat disorders by Southern blot and by PCR sizing
- Are you able to compare and contrast the limitations and advantages of each of the molecular tests
- Explain to your friend how the following techniques are used to understand chromosomal abnormalities: karyotyping, chromosome banding (staining), FISH (fluorescent in situ hybridization), SKY (spectral karyotyping), array CGH (Comparative Genomic Hybridization), array SNP chips
- Compare, contrast, analyze and differentiate between techniques that are able to identify whole chromosome defects (aneuploidy) and chromosomal microdeletions
- Describe how chromosomal analysis can identify:
 - Aneuploidy
 - Mapping of chromosomal breakpoints
 - Detection of subtle translocations
 - Characterization of complex rearrangements
 - Gains, losses and amplifications of DNA
- Can you describe the general concept of NGS (next generation sequencing)
- Discuss with your friend about nucleic acid-based microarrays
 - cDNA microarrays (gene expression at the mRNA level)
 - Array CGH
 - SNP chips
- Compare and contrast the effects of and discuss the various types of mutation that occur in the genome including:
 - substitution (missense, nonsense)
 - insertion/deletion (in-frame, frameshift)
 - splice-site
 - amplification of repeat sequence, genomic instability
 - regulatory mutation (promoter, regulatory element)
- Clinical examples include diseases involving single base mutation (sickle cell anemia), and repeat expansion disorders (Fragile X, Huntington disease, myotonic dystrophy). Other examples include trisomy syndromes such as Down syndrome (trisomy Chr. 21); Microdeletions such as DiGeorge syndrome (22q11.2 - Chr. 22 microdeletion); and cancer (due to genome rearrangements and amplifications).



STUDY
TO
SAVE
LIVES

The End